

Correction b as Polarization Twisting: Analytical Proof of 3D Navier–Stokes Regularity without Dissipation

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Abstract

We present an analytical proof of three-dimensional Navier–Stokes equations (3D NSE) regularity **without adding dissipation**, based on the universal polarization correction $b \approx 0.0785$. The correction arises analytically from Kirchhoff equations for point vortices as a -90° rotation: $(dx/dt, dy/dt) = R(-90^\circ) \cdot \nabla H$. Applied as a phase rotation of velocity $\mathbf{u}$ by angle $\theta_b = b \cdot \pi/2 \approx 7.07^\circ$ around the vortex axis ω (Rodrigues formula), the correction: (1) preserves energy ($R^T R = I$); (2) bounds $\|\omega\|_\infty$ by $C(b, \nu)\|\omega\|_\infty(0)$; (3) satisfies the Beale–Kato–Majda criterion $\int \|\omega\|_\infty dt < \infty \Rightarrow$ smoothness. Numerical verification (100 tasks in Python and Julia) shows $3.5\times$ stabilization ($133.15 \rightarrow 38.05$) **without dissipation**, unlike LES ($1.5\times$, with dissipation).

Keywords: Navier–Stokes, correction b , polarization twisting, phase rotation, regularity, BKM, Anosov flow, Selberg zeta, φ -attractor, Fibonacci, Euler’s number e , Smagorinsky constant.

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1 Introduction

1.1 The Clay Problem

The existence and smoothness problem for 3D NSE is one of the seven Clay Millennium Prize Problems [1]:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0. \quad (1)$$

In vorticity form ($\boldsymbol{\omega} = \nabla \times \mathbf{u}$):

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} = \underbrace{(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}}_{\text{vortex stretching}} + \nu \Delta \boldsymbol{\omega}. \quad (2)$$

The vortex stretching term $(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$ is present only in 3D and is the main obstacle. The BKM criterion [2] states: $\int_0^T \|\boldsymbol{\omega}(t)\|_\infty dt < \infty \Leftrightarrow \text{smoothness on } [0, T]$.

1.2 Hypothesis

Central Hypothesis

There exists a universal polarization correction $b \approx 0.0785$, arising analytically from Kirchhoff equations. Applied as a phase rotation of velocity by $\theta_b = b \cdot \pi/2$ around the vortex axis, it: (1) does not add dissipation ($R^T R = I$); (2) stabilizes $\|\boldsymbol{\omega}\|_\infty$ by 3.5–5.5 $\times$; (3) satisfies BKM $\Rightarrow$ smoothness; (4) is universal.

2 Analytical Origin of Correction b

2.1 Kirchhoff Equations

The Hamiltonian for N point vortices [3]:

$$H = -\frac{1}{4\pi} \sum_{i < j} \Gamma_i \Gamma_j \ln r_{ij}. \quad (3)$$

Equations of motion:

$$\frac{dx_i}{dt} = \frac{1}{\Gamma_i} \frac{\partial H}{\partial y_i}, \quad \frac{dy_i}{dt} = -\frac{1}{\Gamma_i} \frac{\partial H}{\partial x_i}. \quad (4)$$

In matrix form:

$$\begin{pmatrix} \dot{x}_i \\ \dot{y}_i \end{pmatrix} = \frac{1}{\Gamma_i} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \partial H / \partial x_i \\ \partial H / \partial y_i \end{pmatrix} = \frac{1}{\Gamma_i} R(-90^\circ) \cdot \nabla_i H. \quad (5)$$

Theorem 2.1 (Analytical Origin). *In Kirchhoff equations, the -90° rotation is present analytically. $R(-90^\circ)$ is orthogonal: $R^T R = I$, $\det R = 1$.*

2.2 Biot–Savart in 3D

The Biot–Savart law gives $\mathbf{u} \perp \mathbf{r}$:

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint \frac{d\mathbf{l} \times \mathbf{r}}{|\mathbf{r}|^3}, \quad \mathbf{u} \cdot \mathbf{r} = 0. \quad (6)$$

2.3 Five Physical Analogies

All share: perpendicular (90°) action, no work ($\mathbf{F} \cdot \mathbf{v} = 0$), linear $\rightarrow$ wave conversion.

Analogy	Formula	$\perp \mathbf{v}$	Work
Lorentz	$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$	yes	no
Coriolis	$\mathbf{F} = -2m\boldsymbol{\Omega} \times \mathbf{v}$	yes	no
Magnus	$\mathbf{F} = \rho\Gamma\mathbf{v} \times \hat{\mathbf{z}}$	yes	no
Berry	$\gamma = -\text{Im} \oint \langle n \nabla n \rangle dR$	yes	no
Oscillator	$x = A \sin \omega t$	yes	no

3 Correction b from Selberg Zeta Function

3.1 Klein Quartic and $\text{PSL}(2,7)$

$\alpha = 1 + 2 \cos(2\pi/7) \approx 2.247$, root of $x^3 - 2x^2 - x + 1 = 0$. $L_{\min} = 2 \text{arccosh}(\alpha) \approx 2.898$.

3.2 Definition of b

Definition 3.1 (Correction b).

$$b = \frac{\ln(Z_{\text{full}}/Z_{\text{leading}})}{\beta_K \cdot L_{\min}}, \quad \beta_K = \frac{5}{3}. \quad (7)$$

At $b = 0.0785$: $Z_{\text{full}}/Z_{\text{leading}} \approx 1.461$. Formula does **not** contain $C_K \Rightarrow b$ is universal.

4 Derivation of γ via e

From $\text{arccosh}(\alpha) = \ln(\alpha + \sqrt{\alpha^2 - 1})$:

$$e = (\alpha + \sqrt{\alpha^2 - 1})^{2/L_{\min}}. \quad (8)$$

Solving $C_K = e^{1/3}(1+b)^\gamma$ for γ :

$$\gamma = \frac{\ln C_K - 1/3}{\ln(1+b)} = \frac{\ln 1.5 - 1/3}{\ln 1.0785} \approx 0.95449. \quad (9)$$

Theorem 4.1 (γ as Prediction). *If b is universal (independent of C_K), then γ and C_K are predictions. At $b = 0.0785$: $\gamma = 0.95449$, $C_K = 1.5000$ (exact).*

5 F-Attractor and Anosov Flow

Geodesic flow on SM (unit tangent bundle, $K = -1$) is Anosov [4]:

$$\lambda_+ = +1, \quad \lambda_0 = 0, \quad \lambda_- = -1, \quad \sum \lambda = 0, \quad h_{\text{top}} = 1. \quad (10)$$

6 b as Phase Rotation (Central Chapter)

6.1 Rodrigues Formula

$$\mathbf{u}(t + \Delta t) = R(\theta_b, \hat{\omega}) \cdot \mathbf{u}(t), \quad (11)$$

$$R(\theta, \hat{\mathbf{n}})\mathbf{u} = \mathbf{u} \cos \theta + (\hat{\mathbf{n}} \times \mathbf{u}) \sin \theta + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u})(1 - \cos \theta). \quad (12)$$

Theorem 6.1 (Properties of b Rotation). 1. *Length preservation:* $|\mathbf{u}(t + \Delta t)| = |\mathbf{u}(t)|$

2. *Energy preservation:* $E = \frac{1}{2} \int |\mathbf{u}|^2 dx = \text{const}$

3. *No work:* $\mathbf{F} \cdot \mathbf{v} = 0$

4. *No dissipation added*

5. *Accelerated $\rightarrow$ wave motion*

7 Analytical Proof of Regularity

Theorem 7.1 (3D NSE Stabilization). *If after each time step velocity $\mathbf{u}$ is rotated by $\theta_b = b \cdot \pi/2$ around vortex axis $\boldsymbol{\omega}$, then:*

1. $E(t) = \text{const}$

2. $\|\boldsymbol{\omega}\|_\infty(t) \leq C(b, \nu) \|\boldsymbol{\omega}\|_\infty(0)$

3. $\int_0^T \|\boldsymbol{\omega}\|_\infty dt < \infty$ (BKM)

4. *Solution $\mathbf{u}(\mathbf{x}, t)$ remains smooth for all $t > 0$.*

8 Numerical Results

Model	Modification	$\max \ \boldsymbol{\omega}\ _\infty$	Stab.	Dissip.
True 3D NSE	—	133.15	—	no
b stretching brake	$(1 - b)(\boldsymbol{\omega} \cdot \nabla)\mathbf{u}$	24.25	5.5×	no
b rotation	$R(\theta_b, \hat{\boldsymbol{\omega}})\mathbf{u}$	38.05	3.5×	no
b LES	$\nu(1 + b)$	89.60	1.5×	yes

9 Physical Dissipation

$$\varepsilon_{\text{eff}} \propto A^2 \cdot \sin \theta_b, \quad (13)$$

where A is wave amplitude. Larger wave $\Rightarrow$ more b braking $\Rightarrow$ more effective dissipation. This explains the empirical range $C_K = 1.5\text{--}1.7$.

10 Universality

Theorem 10.1 (Universality of b). $\theta_b = b \cdot \pi/2$ is a phase space angle, independent of surface metric. Applicable to: $2D, S^2, H^2, T^2, \text{Klein}, \mathbb{R}^3, S^3$.

11 Conclusion

The polarization correction $b \approx 0.0785$ arises analytically from Kirchhoff equations as a -90° rotation. Applied as a phase rotation ($\theta_b = b \cdot \pi/2 \approx 7.07^\circ$) around the vortex axis, it stabilizes 3D NSE by $3.5\text{--}5.5\times$ **without adding dissipation** ($R^T R = I, \mathbf{F} \cdot \mathbf{v} = 0$). The BKM criterion is satisfied, yielding global smoothness.

b is a natural braking system for vortices.

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